MATHEMATICS

MATERIAL FOR GRADE 12

Differential Calculus

QUESTIONS

QUESTION 1

1.1 Differentiate, using the first Principles, the function $f(x) = -\frac{3}{x}$ (5)

1.2 Differentiate:

1.2.1 $y = \frac{x^2 - 3x + 1}{x^3}$ (4)

1.2.2 $y = \frac{\sqrt{t}}{2} - \frac{4}{7t^5}$ (4)

[13]

QUESTION 2

2.1 Given $f(x) = (x+1)^2(x-2) = x^3 - 3x - 2$ and $g(x) = 3x^2 + 3x$

2.1.1 Calculate the x −intercepts of f and g. (2)

2.1.2 Calculate the co-ordinates of the turning points f and g. (4)

2.1.3 Sketch the graphs showing all the relative turning points and the intercepts with the axes. (6)

2.2 Determine the solution of $2 + 4x + 3x^2 - x^3$ using your graphs. (2)

2.3 For which value(s) of x will $f(x).g(x) < 0$? (3)

[17]

QUESTION 3

3.1 A Petrol tank at BP Depot has both the inlet and the outlet pipes which are used to control the amount of petrol it contains. The depth of the tank is given by $D(t) = 6 + \frac{t^2}{4} - \frac{t^3}{8}$ where D is in metres and t is in hours that are measured from 9h00.

3.1.1 Determine the rate at which the depth is changing at 12h00, and then tell whether there is and increase or decrease in depth. Answer correct to two decimal digits. (3)

3.1.2 At what time will the inflow of petrol be the same as the outflow? (4)

QUESTION 4

4.1 Determine the derivative of $f(x) = 2x^2 - x$ from first principles. (5)

4.2 Determine $\frac{dy}{dx}$ if :

4.2.1 $y = (x + x^{-2})^2$ (4)

4.2.2 $xy^2 = 4$ (3)

4.3 If it is given that $f(x) = x^2 - \frac{4}{x^2}$,

4.3.1 Determine the gradient of the tangent at the point where $x = 2$. (3)

` 4.3.2 Determine the equation of the tangent in question 7.3.1 above. (3)

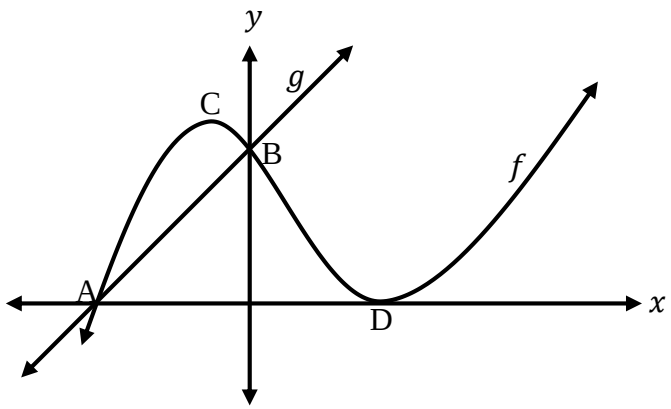
[18]

QUESTION 5

The equations of the graphs below are $f(x) = (x-2)^2(x-a)$ and $g(x) = mx + 12$.

They are both passing through points A and B. C and D are turning points of f .

A and D are the $x-$ intercepts of f. B is the $y-$ intercept of f and g.



5.1 Write down the coordinates of B. (1)

5.2 Determine the coordinates of A and D. (3)

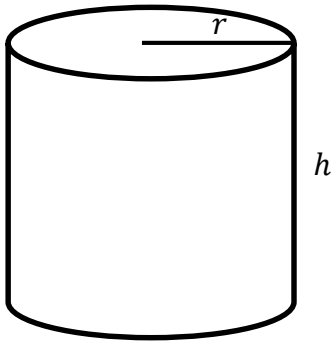
5.3 Determine the coordinates of C. (6)

5.4 For which values of x will f be concave down? (3)

[13]

QUESTION 6

A 340 ml can of cool drink with height h and radius r is shown below.



6.1 Determine the height of the can in terms of the radius r. (3)

6.2 Show that the surface area of the can be written as $SA = 2\pi r^2 + \frac{680}{r}$. (2)

6.3 Determine the radius of the can in cm, if the surface area of the can has to be as small as possible. (4)

[9]

QUESTION 7

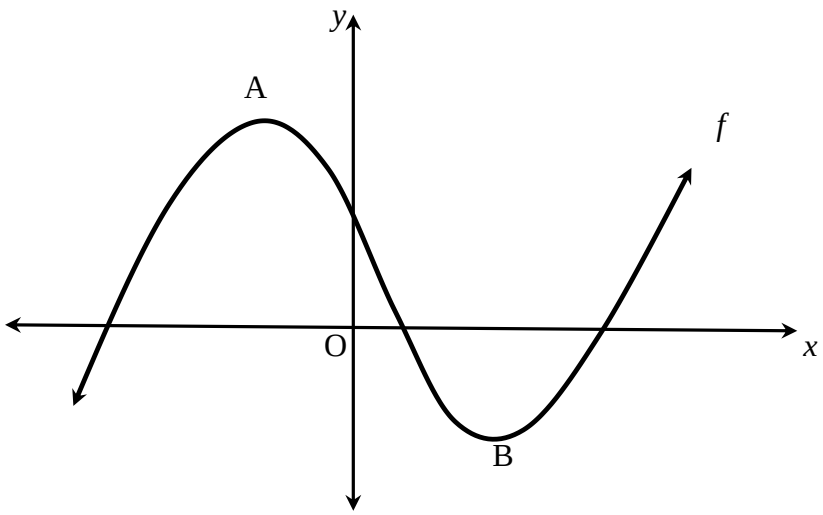
Determine $f'(x)$ from first principles if $f(x) = 3x - x^2$. (5)

Determine $\frac{dy}{dx}$ if $y = \frac{3x}{5x^2} - \frac{1}{2\sqrt{x}}$. (4)

[9]

QUESTION 8

The graph of $f(x) = x^3 - 4x^2 - 11x + 30$ is sketched below. A and B are turning points of *f*.



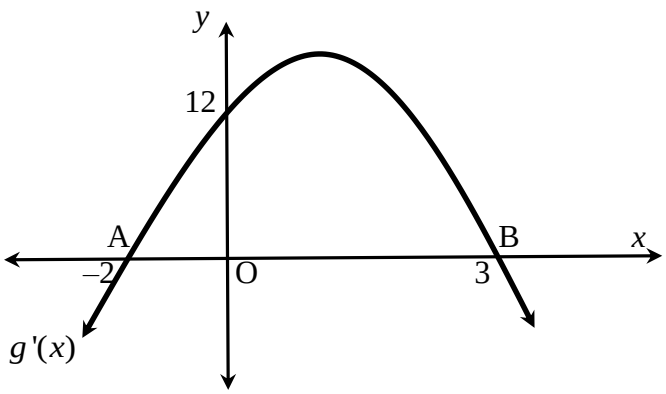
8.1.1 Determine the coordinates of A and B. (6)

8.1.2 Determine the *x*-coordinate of the point of inflection of f. (2)

8.1.3 Determine the equation of the tangent to f at $x = 2$ in the form $y = mx + c$. (4)

8.1.4 Determine the value(s) of *k* for which $x^3 - 4x^2 - 11x + 30 - k = 0$ will have only ONE real root. (2)

The graph of $y = g'(x)$ is sketched below, with x-intercepts at A $(-2; 0)$ and B $(3; 0)$. The y-intercept of the sketched graph is $(0; 12)$.



8.2.1 Determine the gradient of g at $x = 0$. (1)

8.2.2 For which value of x will the gradient of g be the same as the gradient in QUESTION 8.2.1? (1)

8.2.3 Draw a sketch graph of $y = g(x)$. Show the x-values of the stationary points and the point of inflection on your sketch. It is not necessary to indicate the intercepts with the axes. (3)

[19]

QUESTION 9

A stone is thrown vertically upwards. Its height (in metres) above the ground at time t (in seconds) is given by: $h(t) = -2t^2 + 12t + 32$.

9.1 Determine the initial height of the stone above the ground. (1)

9.2 Determine the time taken to reach the maximum height. (3)

9.3 How fast was the stone travelling when it hit the ground? (5)

9.4 Determine the acceleration of the stone. (1)

[10]

QUESTION 10

10.1 If $f(x) = -2x^2$, determine $f'(x)$ from first principles. (5)

10.2 Determine:

10.2.1 $\frac{dy}{dx}$ if $y = \frac{2x^2 - 1}{\sqrt{x}}$ (3)

10.2.2 $D_x[(3x - 2)^2]$ (3)

10.3 Given: $y = \frac{1}{x^2}$.

Prove that the gradient of the curve is negative at each point on the curve where $x > 0$. (3)

[14]

QUESTION 11

The following information about a cubic polynomial, $y = f(x)$, is given:

- $f(-1) = 0$
- $f(2) = 0$
- $f(1) = -4$
- $f(0) = -2$
- $f'(-1) = f'(1) = 0$
- if $f(x) < -1$ then $f'(x) > 0$
- if $f(x) > 1$ then $f'(x) > 0$

11.1 Use this information to draw a neat sketch graph of f using the grid on the DIAGRAM SHEET (5)

11.2 For which value(s) of x is f strictly decreasing? (2)

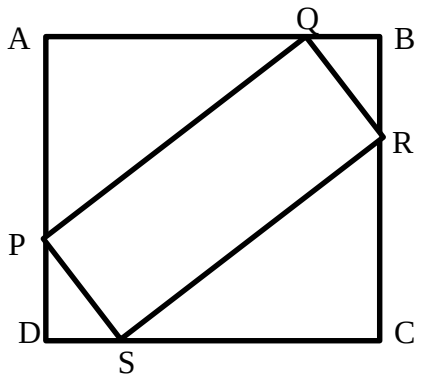
11.3 Use your graph to determine the x-coordinate of the point of inflection of f. (2)

11.4 For which value(s) of x is f concave up? (2)

[11]

QUESTION 12

ABCD is a square with sides 20 mm each. PQRS is a rectangle that fits inside the square such that $QB = BR = DS = DP = k$ mm.



12.1 Prove that the area of $PQRS = -2k(k - 20) = 40k - 2k^2$ (4)

12.2 Determine the value of k for which the area of $PQRS$ is a maximum. (4)

[8]

QUESTION 13

13.1 Determine $f'(x)$ from first principles if $f(x) = x^3$ (5)

13.2 Determine:

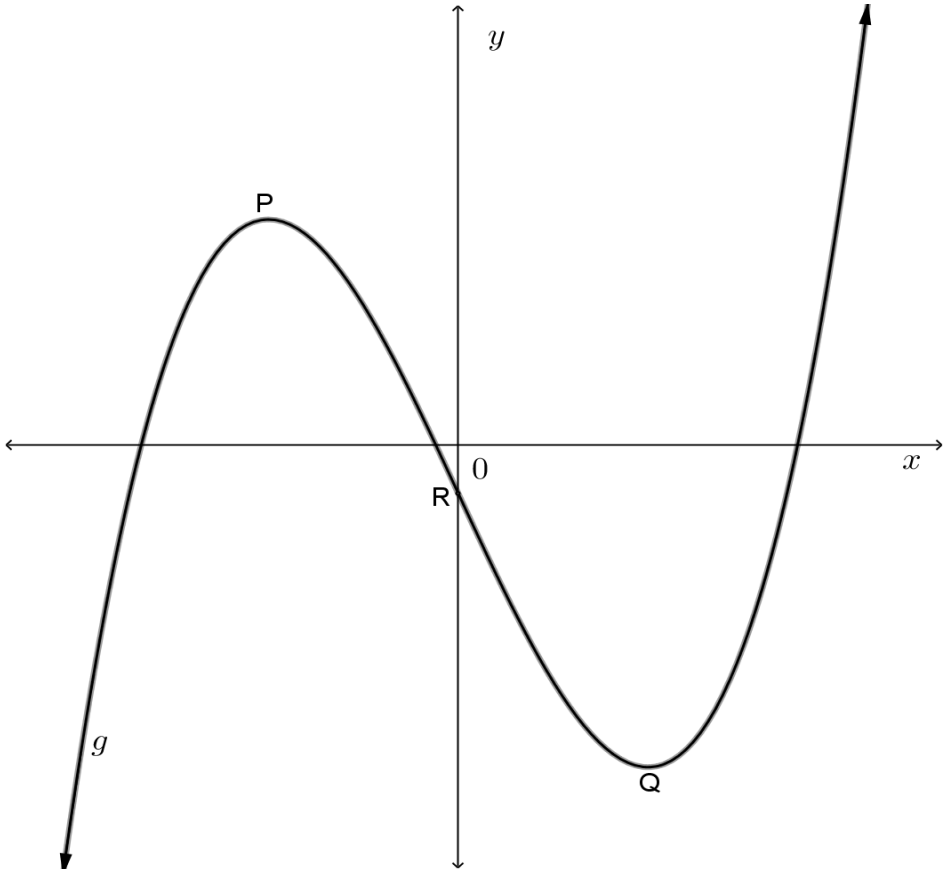
13.2.1 $f'(x)$ if $f(x) = \left(x + \frac{1}{x}\right)^2$ (3)

13.2.2 if $y = \frac{3}{x^{\frac{2}{3}}} - \pi x$ (3)

[11]

QUESTION 14

The sketch below shows the graph of $g(x) = x^3 - 6x - 1$. P and Q are the turning points and R the *y*-intercept of *g*.



14.1 Determine the coordinates of R. (2)

14.2 Determine the coordinates of the turning points P and Q. (6)

14.3 Calculate the values of x for which g strictly increases as x increases. (2)

14.4 If $h(x) = g'(x)$, determine for which values of x is $h(x) \leq 0$. (2)

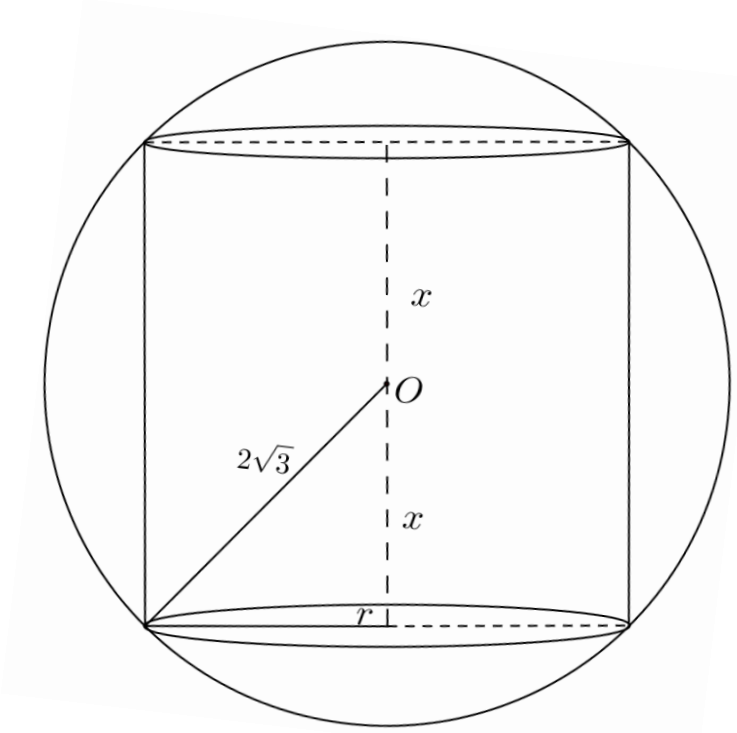
14.5 Determine the equation of the tangent to *g* at R. (4)

14.6 Write down the equation of the line perpendicular to the tangent at P. (2)

[18]

QUESTION 15

The diagram below shows a cylindrical can that fits into circle O with radius $R = 2\sqrt{3}$ units.



15.1 Express r in terms of x. (2)

15.2 Calculate the value of x for which the volume of the cylinder is a maximum. (5)

15.3 Calculate the maximum volume of the can in terms of π. (2)

[9]

QUESTION 16

16.1 Given $f(x) = x^3$. Determine the derivative of f from first principles. (5)

16.2 Determine:

16.2.1 $f'(x)$ if $f(x) = \sqrt{x} - \frac{4}{x^2}$ (4)

16.2.2 $\frac{dy}{dx}$ if $\frac{1}{2}x^2 - 3 = \sqrt{y}$ (4)

[13]

QUESTION 17

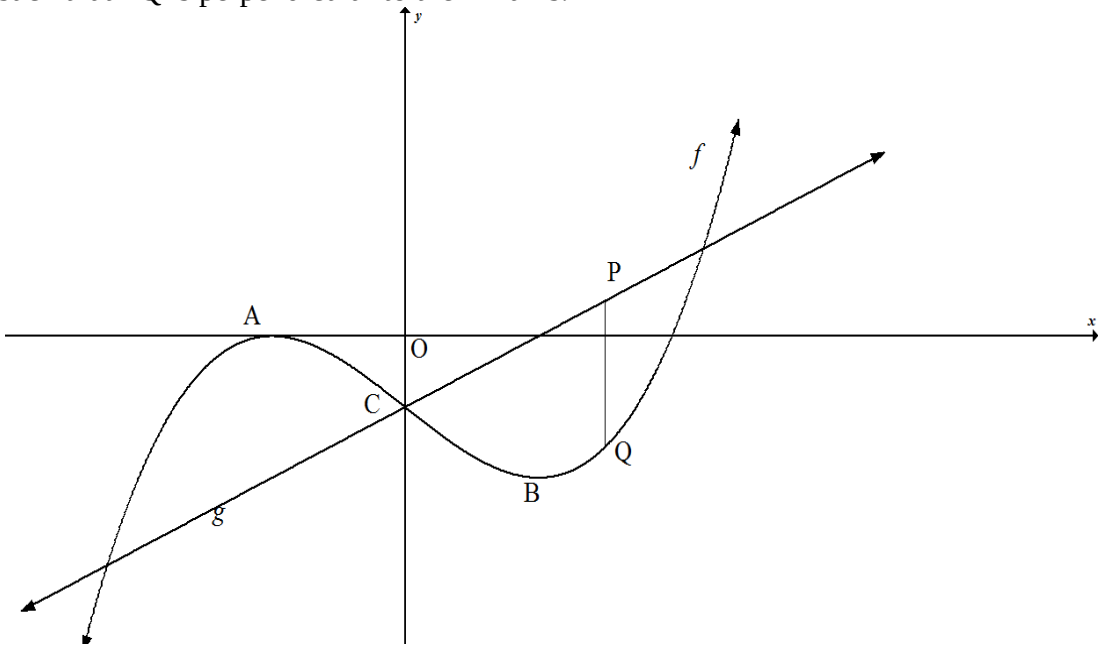
17.1 Given $f(x) = x^2 - 8$

17.1.1 Calculate $f(-3)$. (1)

17.1.2 Calculate $f'(-3)$. (1)

17.1.3 Determine the equation of the tangent to $f(x) = x^2 - 8$ at $x = -3$. (2)

17.2 The graph of a cubic function with equation $f(x) = x^3 - 3x - 2$ and $g(x) = 2x - 2$ is drawn. A and B are the turning points of f. P is a point on g and Q is a point on f such that PQ is perpendicular to the x – axis.



17.2.1 Calculate the coordinates of A and B. (4)

17.2.2 If PQ is perpendicular to the x – axis, calculate the maximum length of PQ, (4)

17.2.3 Determine the values of k for which $f(x) = k$ has only two real roots. (2)

17.2.4 Determine the values of x for which f is concave up. (3)

[17]

QUESTION 18

A car speeds along a 1 kilometre track in 25 seconds. Its distance (in metres) from the start after *t* seconds is given by

$s(t) = t^2 + 15t$

18.1 Write down an expression for the speed (the rate of change of distance with respect to time) of the car after *t* seconds. (1)

18.2 Determine the speed of the car when it crosses the finish line. (1)

18.3 Write down an expression for the acceleration (the rate of change of speed with respect to time) of the car after *t* seconds. (2)

18.4 Hence, or otherwise, calculate the acceleration of the car after 5 seconds. (1)

18.5 Calculate the speed of the car when it is 250 metres down the track. (4)

[9]

QUESTION 19

19.1 Determine the derivative of $f(x) = 2x - x^2$ from first principle. (5)

19.2 Differentiate the following with respect to x:

19.2.1 $f(x) = 6x^4 - \sqrt[3]{x^{-2}}$ (3)

19.2.2 $y = (4x^2)^3 + 2x$ (3)

[11]

QUESTION 20

Given: $f(x) = ax^3 + bx^2 + 3x + 3$ and $g(x) = f''(x)$ where $g(x) = 12x + 4$.

20.1 Show that $a = 2$ and $b = 2$. (4)

20.2 Prove that f will never decrease for any real value of x. (5)

20.3 Determine the minimum gradient of f. (4)

20.4 Explain the concavity of f for all values of x where $g(x) < 0$. (1)

[14]

QUESTION 21

A particle moves according to the function $H(t) = -2t^2 + 70t$, where t is time in seconds and H is height in meters above the ground.

21.1 Calculate the height of the particle after 5 seconds. (2)

21.2 How much time did it take for the particle to strike the ground again? (4)

21.3 What was the maximum height that the particle reached? (4)

21.4 What was the initial speed (velocity) of the particle? (2)

[12]

QUESTION 22

Determine $f'(x)$ from first principles if $f(x) = \frac{3}{x}$. (5)

Determine $\frac{dy}{dx}$ if:

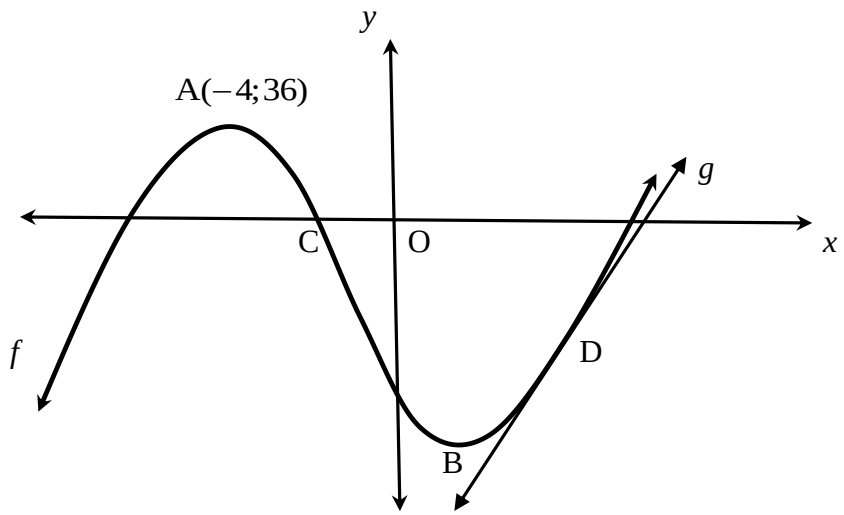
22.2.1 $y = \pi^3 x - \sqrt[3]{x}$ (3)

22.2.2 $y = \frac{7x^5 - 3x}{4x}$ (2)

[10]

QUESTION 23

The function defined by $f(x) = x^3 + px^2 + qx - 12$ is sketched below. A$(-4; 36)$ and B are turning points of *f*. *g* is a tangent to the graph of *f* at D.



23.1 Show that $p = 5$ and $q = -8$. (6)

23.2 If C$(-1; 0)$ is an *x*-intercept of *f*, calculate the other *x*-intercepts of *f*. (4)

23.3 Determine the equation of g, the tangent to *f* at D (1; – 14). (4)

23.4 For which values of k will $f(x) = g(x) + k$ have two positive roots? (2)

[16]

QUESTION 24

A marathon athlete trains between two towns A and C. He starts at point B which lies between towns A and C. The athlete runs from point B to town C and back to point B. The road between the towns is in a straight line. The displacement S, in kilometres, from point B after *t* hours, is given by:

$S(t) = -t^3 + 12t^2 - 32t$

24.1 How many hours will it take the athlete to return to point B? (3)

24.2 Calculate the distance between point B and town C. (5)

24.3 Calculate the maximum speed that the athlete has reached while training. (4)

[12]

QUESTION 25

25.1 Determine $f'(x)$ using first principles if $f(x) = \frac{2}{x}$. (5)

25.2 Determine the following:

25.2.1 $h'(x)$ if $h(x) = 2x^3 - \frac{4}{x} + 3\sqrt{x}$ (4)

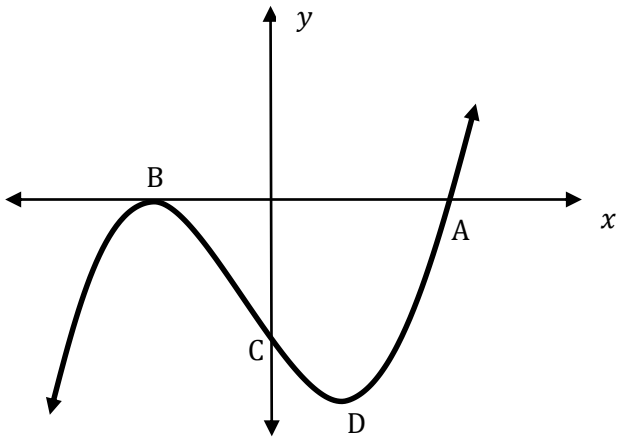
25.2.2 $D_x\left(\frac{8x^3 - 27}{2x - 3}\right)$ (3)

25.3 Explain, by doing the necessary calculations, why the tangent to the curve of $f(x) = -x^3 - 2x$ will never have a positive gradient. (3)

[15]

QUESTION 26

The diagram below shows the graph of $f(x) = x^3 + x^2 - x - 1$.



26.1 Calculate the distance between A and B, the *x*-intercepts. (5)

26.2 Calculate the coordinates of D, a turning point of f. (3)

26.3 Show that the concavity of f changes at $x = -\frac{1}{3}$ (3)

26.4 For which values of x is:

26.4.1 $f(x) > 0$ (1)

26.4.2 $f(x).f'(x) < 0$ (3)

[15]

QUESTION 27

A water tank with an inlet and an outlet is used to water a garden. The equation $D = 3 + \frac{1}{2}t^2 - \frac{1}{4}t^3$ gives the depth of water in metres where t is the time in hours that has elapsed since 09:00.

27.1 What is the depth of the water at 11:00 (2 hours later)? (1)

27.2 At what rate does the depth of the water change at 12:00? (2)

27.3 At what time will the inflow of the water be the same as the outflow of water? (4)

[7]